Akoizo Original Practice - Form 5 - Answer key

Division C | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which is quantitative evidence?

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

A. Means increased from 12 to 20 units

B. It looks nice

C. The hypothesis is obviously true

D. The team worked hard

Correct: A

2. (2 points) Randomizing order mainly helps address…

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

A. Every possible confounder automatically

B. Time-linked systematic differences

C. All measurement uncertainty

D. The need for units

Correct: B

3. (2 points) 2.0 x 3.00 cm^2 is reported as…

A. 6.00000 cm^2

B. 6 cm

C. 5.0 cm^2

D. 6.0 cm^2

Correct: D

4. (2 points) High R^2 proves causation?

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

A. No

B. Yes

C. Only with a colored line

D. Only when n = 3

Correct: A

5. (4 points) Name the example independent and dependent variables.

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

answer: Independent: release height. Dependent: transit time between defined marks. (4 points).

6. (4 points) Why do more repeats not remove a fixed offset?

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

answer: The mean retains the systematic shift even if random uncertainty decreases. (4 points).

7. (4 points) Why separate raw and summary data?

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

answer: It permits auditing and recalculation without confusing original observations with derived quantities. (4 points).

8. (4 points) How can roles create contradictions?

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

answer: If teammates use different units, endpoints, variable names, or procedure versions without coordination. (4 points).

9. (4 points) Why show with/without-outlier results?

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

answer: It reveals whether the conclusion depends strongly on the flagged value under transparent alternatives. (4 points).

10. (4 points) Why does replication not solve drift alone?

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

answer: More trials can reproduce the same systematic level-time distortion if order remains confounded. (4 points).

11. (4 points) Why check the applicable checklist?

Unit: Division C comparison standards, significant figures, and abstracts.
Worked-case data: Report area from measured lengths 2.0 cm and 3.00 cm, then report their sum.

answer: Required sections, competition-level applicability, and scoring can change; the syllabus teaches skills rather than fixing current rules. (4 points).

12. (6 points) Solve this case and explain your reasoning: For response values 8, 10, 12, calculate mean, range, and sample SD.

reasoning: Mean = 30/3 = 10. Range = 12 - 8 = 4.
Squared deviations are 4, 0, 4; sum = 8.
Sample variance = 8/(3-1) = 4; sample SD = 2. (3 points).

conclusion: Mean 10, range 4, and sample SD 2 have distinct meanings; variance is 4 squared response units. (3 points).

13. (2 points) Define “Independent variable” in the context of Experimental Design.

answer: Factor deliberately changed. (2 points).

14. (2 points) Define “Random error” in the context of Experimental Design.

answer: Unpredictable variation across observations. (2 points).

15. (2 points) Define “Raw data” in the context of Experimental Design.

answer: Original recorded measurements before summary processing. (2 points).

16. (2 points) Define “Coherence” in the context of Experimental Design.

answer: Agreement among design, data, analysis, and conclusions. (2 points).

17. (2 points) Define “Reasoning” in the context of Experimental Design.

answer: Explanation connecting evidence to a claim. (2 points).

18. (2 points) Define “Confounding” in the context of Experimental Design.

answer: A competing factor systematically linked with the intended factor. (2 points).